

Ziwei Ma

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Department of Mathematics
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Education

New Mexico State	Las Cruces, NM	Jan 2014 – May 2020
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University

- **Ph.D. in Mathematics (Concentration: Mathematical Statistics)**

Advisor: Prof. Tonghui Wang *Thesis title: “Estimation of Parameters and Decomposition of Quadratic Forms Under Skew-Normal Settings”*

- **M.S. in Mathematics**

Northwest University	Xi’an, China	Aug 2000 – Jul 2007
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- **M.S. in Mathematics**
- **B.S. in Mathematics**

Professional Experience

Assistant Professor	University of Tennessee at Chattanooga	Aug 2020 – Present
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Biostatistician	University of Tennessee College of Medicine - Chattanooga	Aug 2021 - Present
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Teaching/Research Assistant	New Mexico State University	Jan 2014 - May 2020
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Research Interests

Statistics

- Linear Model, Skew Normal Distribution, Inferential Models, Clinic Trials

Applied Math/Stats

- Machine Learning in Engineering, Applications of Stochastic Differentiation Equations in Biology and Medical Science

Grants

1. **PI** Biomedical Research Initiation Collaborative grant (BRIC), sponsored by Erlanger-UTCOMUTC Collaborative Grant Program, “Prevalence and Disease Characteristics of Alpha-1 Antitrypsin Deficiency (AAT) in a Southeastern United States Patient Population and its Association with Chronic Obstructive Pulmonary Disease (COPD), Pulmonary fibrosis, Hospital Readmission, and Mortality” (**\$ 25,000, Jan 2023 - Jun 2024**)
my role: Conducting data cleaning, develop statistical methods, prepare for the report, presentation, and manuscript. Four poster presentations in national and regional conference, one manuscript is in preparation. Supervised graduate student Opoku Foster (enrolled in MS program in Fall 2023) who is supported by this grant from Fall 2023 to Spring 2024. Also, MS students Andrew Stuckey and Hagar Cobbinah were supported.
2. **Co-PI** NSF (NSF # 2200138, PI: Dr. Hong Qin from Computer Science at UTC) “PIPP Phase I: Develop and Evaluate Computational Frameworks to Predict and Prevent Future Coronavirus Pandemics” (**\$ 1,000,000, Aug 2022 - Jan 2026**)
my role: Provide statistical consulting, conduct data analysis, leading resampling process for further modeling
3. **Co-PI** CEACSE grant sponsored by Center of Excellence in Applied Computational Science and Engineering at UTC (PI: Dr. Lani Gao from Math department at UTC) “Integrating google trends analytic into geographically weighted model of vaccine hesitancy” (**\$ 95,577, Jul 2022 - Jun 2023**)
my role: Collecting data, develop statistical model, mentoring graduate students
4. **Faculty mentor** NSF REU grant (NSF#1852042, PI: Dr. Hong Qin from Computer Science at UTC) REU Site: Interdisciplinary Computational Biology (**\$ 413,526, Mar 2022 - Feb 2026**)
my role: Mentor undergraduate students for 10 weeks in summer. In summer 2021, I mentored Landon Kehr from Taylor University on project “Addressing sampling biases in genome-wide association study for SAR-CoV-2”, and Daniel Bosah from Bronx Community College of the City University of New York on project “mortality pattern of yeast RLS because he is the only one putting first choice on this project”
5. **Co-PI** CEACSE grant sponsored by Center of Excellence in Applied Computational Science and Engineering at UTC (PI: Dr. Hong Qin from Computer Science at UTC) Addressing sampling biases in genome-wide association study for SAR-CoV-2 (**\$ 90,000, Jul 2021 - Jun 2022**)
my role: Collecting data, develop statistical models, explore resampling techniques, mentoring students. Gertrude Osei(enrolled in MS program in Fall 2021) was supported in Spring 2022.
6. **Co-I** NIH (NIH # NIH 3R15GM131315-01A1S2 WANG 19-22, PI: Dr. Jin Wang from Math department at UTC) A computational modeling framework for COVID-19 vaccination (**\$ 122,580, Jun 2021 - May 2022**)
my role: Conducting statistical analysis and exploring machine learning algorithms.

Honors/Awards

- **NSF QIP Mathematical Fellowship 2025** Partially supporting travel to attend QIP 2025
- **UTC Math Travel Grant** Partially supporting travel to attend QIP 2025
- **NSF SIAM Quantum Intersections Convening, Oct. 2024** Supporting travel to attend the workshop
- **UTC College of Arts and Sciences Graduate Engagement Award** Offered one half-time (10 hour/week) Graduate Assistantship for 2024 - 2025 academic year
- **UTC Math Travel Grant** Supporting travel for Alamo Symposium in Statistics at The University of Texas at San Antonio in March 2023
- **Dean's Graduate Award for Excellence of College of Arts and Sciences, Dec. 2019** The highest award given to a graduate student from the College of Arts and Sciences at NMSU
- **Outstanding Graduate Student 2018 -2019** at NMSU
- **ASA Joint Research Conference Student Travel Award 2018** at Santa Fe, NM
- **ASA Joint Research Conference Student Scholarship 2018** at Santa Fe, NM

Publications

Journal/**B**ook Chapter/**C**onference/**P**reprint, Full list in Google scholar profile

2025

1. [J] Raine, J. D., Rippey, M. G., Miles, M. V. P., **Ziwei Ma**, Carter, B. L., and Bell, C. (2025). *“Re-Evaluating the Necessity of Empiric Anti-Fungal Coverage in Gastroduodenal Perforations”*. The American Surgeon™, 00031348251367025.

2024

2. [J] Xiaonan Zhu, Zheng Wei, Tonghui Wang, S. T. Choy, and **Ziwei Ma**. *“An expectation conditional maximization algorithm for the skew-normal based stochastic frontier model”* Computational Statistics 39, no. 3 (2024): 1539-1558.
3. [C] Shah, R., **Ziwei Ma** and Koo, P., 2024. *“Association Of Heterozygotes For The Alpha-1-antitrypsin Gene With Cirrhosis And Mortality”*. Chest, 166(4), p.A6453.
4. [C] N. Brott, G. Bose, I. Hembree, **Ziwei Ma**, R. Shah, and P. Koo. *“Effect of Heterozygote Alpha-1-antitrypsin Gene on the Development of Pulmonary Fibrosis and Mortality (abstract)”*. Am J Respir Crit Care Med 2024;209:A2112.

2023

5. [B] **Ziwei Ma**, Tonghui Wang, ST Boris Choy, Zheng Wei, and Xiaonan Zhu. *“Extending the A Priori Procedure for Estimating Location Parameter Under Multivariate Skew Normal Settings”* In Optimal Transport Statistics for Economics and Related Topics, pp. 107-116. Cham: Springer Nature Switzerland, 2023.

2022

6. [P] Yifan Hao, Huiping Cao, K. Selcuk Candan, Jiefei Liu, Huiying Chen, and **Ziwei Ma**. “*Class-Specific Attention (CSA) for Time-Series Classification*” arXiv preprint arXiv:2211.10609 (2022).
7. [C] Gresham, Brandon, Juan Torres, Jonathan Britton, **Ziwei Ma**, Anita B. Parada, Michelle L. Gutierrez, Mark Lawrence, and Wei Tang. “*High-dimensional Time-series Gait Analysis using a Full-body Wireless Wearable Motion Sensing System and Convolutional Neural Network*” In 2022 IEEE Biomedical Circuits and Systems Conference (BioCAS), pp. 389-393. IEEE, 2022.
8. [J] **Ziwei Ma**, Ying-Ju Chen, Tonghui Wang, and Jing Liu. “*Inferences on location parameter in multivariate skew-normal family with unknown scale parameter*” Communications in Statistics-Simulation and Computation 51, no. 9 (2022): 5465-5481.
9. [J] **Ziwei Ma**, Tonghui Wang, Zheng Wei, and Xiaonan Zhu. “*Inferences on location parameters based on independent multivariate skew normal distributions*” Asian Journal of Economics and Banking 6, no. 2 (2022): 270-281.

2021

10. [B] **Ziwei Ma**, Wang, Tonghui, Li, Baokun, Xiaonan Zhu and yuede Ma. “*The Decomposition of Quadratic Forms Under Matrix Variate Skew-Normal Distribution*” In Behavioral Predictive Modeling in Economics (pp. 181-194). Springer, Cham. (2021)
11. [B] Li, Baokun, **Ziwei Ma**, and Tonghui Wang. “*Find Trade Patterns in China’s Stock Markets Using Data Mining Agents*” Behavioral Predictive Modeling in Economics (2021): 171-179.

2020

12. [J] Hanwan Jiang, Hanyu Zhan, **Ziwei Ma**, and Ruinian Jiang. “*Comparative Study of Three-Dimensional Stress and Crack Imaging in Concrete by Application of Inverse Algorithms to Coda Wave Measurements*” Sensors 20, no. 17 (2020): 4899.
13. [J] Lei Mu, Zhe Jia, **Ziwei Ma**, Fuhui Shen, Yuekuo Sun, and Yong Zang. “*A theoretical prediction framework for the construction of a fracture forming limit curve accounting for fracture pattern transition*” International Journal of Plasticity 129 (2020): 102706.
14. [J] **Ziwei Ma**, Ben Niu, Tuan Anh Phan, Anne Line Stensjøen, Chibawanye Ene, Timothy Woodiwiss, Tonghui Wang, Philip K. Maini, Eric C. Holland, and Jianjun Paul Tian. “*Stochastic growth pattern of untreated human glioblastomas predicts the survival time for patients*” Scientific Reports 10, no. 1 (2020): 6642.

2019

15. [J] Xiaochen Tang, **Ziwei Ma**, Qisong Hu, and Wei Tang. “*A real-time arrhythmia heartbeats classification algorithm using parallel delta modulations and rotated linear-kernel support vector machines*” IEEE Transactions on Biomedical Engineering 67, no. 4 (2019): 978-986.
16. [J] **Ziwei Ma**, Jinlei Hao, and Tonghui Wang. “*The plausibility regions for shape parameter under multivariate skew-normal settings*” International Journal of Intelligent Technologies & Applied Statistics 12, no. 1 (2019): 31-47.

2018

17. [B] **Ziwei Ma**, Ying-Ju Chen, Tonghui Wang, and Wuzhen Peng. “*The inference on the location parameters under multivariate skew normal settings*” In International Econometric Conference of Vietnam, pp. 146-162. Cham: Springer International Publishing, 2018.

2017

18. [B] **Ziwei Ma**, Xiaonan Zhu, Tonghui Wang, and Kittawit Autchariyapanitkul. “*Joint plausibility regions for parameters of skew normal family*” In International Conference of the Thailand Econometrics Society, pp. 233-245. Cham: Springer International Publishing, 2017.
19. [B] **Ziwei Ma**, Weizhong Tian, Baokun Li, and Tonghui Wang. “*The decomposition of quadratic forms under skew normal settings*” In the International conference of the Thailand econometrics society, pp. 222-232. Cham: Springer International Publishing, 2017.
20. [B] Xiaonan Zhu, **Ziwei Ma**, Tonghui Wang, and Teerawut Teetranont. “*Plausibility regions on the skewness parameter of skew normal distributions based on inferential models*” Robustness in econometrics (2017): 267-286.

2014

21. [J] Junli Du, Zhifa Yuan, **Ziwei Ma**, Jiuzhou Song, Xiaoli Xie, and Yulin Chen. “*KEGG-PATH: Kyoto encyclopedia of genes and genomes-based pathway analysis using a path analysis model*” Molecular BioSystems 10, no. 9 (2014): 2441-2447.

Pending Proposals and Manuscripts

- **Proposal as PI** “Collaborative Research: SCH:Rehabilitation Evaluation of Neurologic Motor Disorders using Wearable Sensors ” (collaboration with New Mexico State University and University of Texas at El Paso), submitted to NSF in October 2025.
- **Proposal as mentor faculty** “REU Site: Interdisciplinary Computational Biology (iCompBio)”, submitted to NSF in August 2025.
- **Manuscript** (in preparation) “The development and application of Skew F distribution”
- **Manuscript** (in preparation) “The skew-normal based Stochastic Frontier Model for panel data”

Invited Talks/Presentations

- **NMSU Math Colloquium Talk**
“*The Development and Applications of Skew F Distribution*” at New Mexico State University in Apr. 2025
- **Alamo Symposium in Statistics**
“*Inference on Multivariate SkewNormal Family and Related Distributions*” at The University of Texas at San Antonio in Mar. 2023

- **International Forum on Collaborative Development of Machine Learning and Digital Economy (virtual)**
“An application of Support Vector Machine (SVM) in Arrhythmia Heartbeats Classification” at Xi'an Technological University, Xi'an China in Dec. 2022
- **AMS Southeastern Sectional Meeting**
“A feed-forward neural network approach to Estimate parameters of models of COVID-19 Dynamics” at UTC in Oct. 2022
- **New England Statistics Symposium (virtual)**
“Estimation in skew Gaussian graphical model” at the University of Connecticut in Nov. 2018
- **New Mexico Tech Math Colloquium Talk**
“The decomposition of quadratic forms under skew normal settings” at UTC in Oct. 2022
- **Joint NMSU/UTEP Workshop** at NMSU or UTEP 11 times from Fall 2014 to Fall 2019

Contributed Talks/Presentations

- **JMM 2020** Joint Mathematics Meetings *“Inference on the difference of location parameters under Multivariate skew-normal setting”* Denver, CO
- **GRAS 2019** Graduate Research & Arts Symposium (GRAS) *“The plausibility regions for shape parameters under multivariate skew normal settings based on Inferential Model”* NMSU, Las Cruces, NM
- **JRC 2018** Joint Research Conference on Statistics in Quality, Industry, and Technology *“Inferences on the location parameter under multivariate skew normal settings”* Sante Fe, NM
- **JMM 2018** Joint Mathematics Meeting *“Inferences on the location parameter under multivariate skew normal settings”* San Diego, CA

Poster Presentation

- **2025 Southeastern Surgical Congress SESC Annual Meeting, New Orleans, LA**
Title: *Re-evaluating the Necessity of Empiric Anti Fungal Coverage in Gastroduodenal Perforations*
Presenter: Raine J (Residents from UTCOM)
- **2024 American College of Chest Physicians CHEST Annual Meeting, Boston, MA**
Title: *Association of Heterozygotes for the Alpha-1-antitrypsin Gene With Cirrhosis and Mortality*
Presenter: Dr. Patrick Koo(Faculty from UTCOM)
- **2024 Southern Medical Association Virtual Abstract Competition**
Title: *Association of A1AT Heterozygotes with Pulmonary Fibrosis and Mortality*
Presenter: Isaac Hembree (Residents from UTCOM)
 Selected in the Top 10 Poster at the Annual Scientific Assembly and won the First place in Poster Competition
- **2024 American Thoracic Society International Conference, San Diego, CA**
Title: *Effect of Heterozygote Alpha-1-antitrypsin Gene on the Development of Pulmonary Fibrosis and Mortality*

Presenter: Nathan Brott (Residents from UTCOM)

- **2024 UTC Fall Research Conference**

Presenter: *Risk Analysis of A1at Heterozygotes with Pulmonary Hospitalization, Readmission, Fibrosis, and Mortality*

Presenter: Hagar Cobbinah (MS student)

- **2024 The 9th Workshop on Biostatistics and Bioinformatics, Atlanta, GA**

Title: *Prevalence of PI*Z and PI*S alleles of alpha-1-antitrypsin deficiency in Tennessee: A Study on Hardy-Weinberg Equilibrium*

Presenter: Foster Opoku (MS student)

- **2024 UTC Spring Research and Arts Conference**

Title: *Prevalence of PI*Z and PI*S alleles of alpha-1-antitrypsin deficiency in Tennessee: A Study on Hardy-Weinberg Equilibrium.*

Presenter: Foster Opoku (MS student)

- **JSM 2019 Joint Statistical Meetings, Denver, CO**

Title: *The decomposition of quadratic forms under matrix variate skew normal Distribution*

Presenter: Ziwei Ma

Seminar/Colloquium

- **2025 UTC Mathematics Colloquium Series** “Some work on the “Wrong skewness” problem in the stochastic frontier model” UTC
- **2024 UTC Mathematics Colloquium Series** “The development and application of generalized F distribution” UTC
- **2023 UTC Mathematics Colloquium Series** “Estimation for Skew-normal based stochastic frontier model” UTC
- **2022 NMSU statistics group Seminar (Zoom)** “Inferences on Location Parameters Based on Independent Multivariate Skew Normal Distributions” NMSU
- **2022 UTC Mathematics Colloquium Series** “An application of Support Vector Machine (SVM) in Arrhythmia Heartbeats Classification” UTC
- **2021 UTC Mathematics Colloquium Series** “A Stochastic Gompertz Diffusion Model for Untreated Human Glioblastomas” UTC
- **2021 NMSU statistics group Seminar (Zoom)** “Estimation of Parameters and Decomposition of Quadratic Forms Under Skew-normal Settings” NMSU
- **2019 NMSU Machine Learning Seminar** “Prototype Methods and Nearest-Neighbors” NMSU
- **2018 NMSU statistics group Seminar** “Skew normality test” NMSU
- **2018 NMSU Bio-math Seminar** “Summer research presentation, Building an SDE model for breast cancer” NMSU
- **2017 NMSU Bio-math Seminar** “Stochastic Differential Equations – Stochastic growth pattern of untreated human glioblastomas predicts the optimal time for surgery” NMSU
- **2016 NMSU statistics group Seminar** “Skew symmetric distributions” NMSU

Teaching

Teaching at UTC

2020 - present

- MATH 2100: Introductory Statistics (one or two sessions every semester since Fall 2020)
- MATH 4130: Introduction to Probability and Statistics (Fall 2021, 2022)
- MATH 4140: Mathematical Statistics (Spring 2022, 2023, 2024)
- MATH 4170: Nonparametric Statistics (Spring 2021)
- MATH 5130: Introduction to Probability and Statistics (Fall 2021, 2023, 2024, 2025)
- MATH 5140: Mathematical Statistics Spring (2022, 2023)
- MATH 5150: Introduction to Biostatistics (Fall 2022)
- MATH 5170: Nonparametric Statistics (Spring 2021)
- MATH 5180: Analysis of Variance and Design of Experiments (Fall 2023)

Teaching at NMSU

2015 - 2020

- All math courses, eligible for graduate students teaching, were taught at least once, such as College Algebra, Intermediate Algebra, Pre-Calculus and Trigonometry, Calculus I, Calculus II, Statistics for Business and Behavior Sciences, Probability and Statistics for Engineers

Advising Students

Students Mentored

** with financial support*

- (Spring 2025) Mentored graduate student (Benedicta Puplampu*)
- (Spring 2025) Mentored undergraduate student (Reid Wilson*)
- (Fall 2024) Mentored graduate student (Benedicta Puplampu*)
- (Summer 2024) Mentored graduate students (Foster Opoku* and Hagar Cobbinah*)
- (Spring 2024) Mentored graduate student (Foster Opoku*)
- (Summer 2023) Mentored graduate student (Andrew Stuckey*)
- (Spring 2023) Mentored graduate student (Gertrude Osei*)
- (Summer 2021) Mentored undergraduates (Landon Kehr from Taylor University and Bosah Daniel from Bronx Community College of the City University of New York) in REU

Graduate Committee

- Chathuri Aththanayake (Ph.D., graduated in Summer 2025)
- Opoku Foster (Master, graduated in Spring 2025)
- Megan McCoy (Ph.D., in progress)
- Gertrude Osei (Master, graduated in Summer 2023)

Recommender

- Wrote recommendation letters for undergraduates and graduates students to graduate schools and scholarships

Conference Organizer/Chair

- **Co-organizer** BCBST Actuarial Fall Visit in 2023
- **Session Chair** in International Forum on Collaborative Development of Machine Learning and Digital Economy (virtual) at Xi'an Technological University, Xi'an China in Dec. 2022
- **Co-organizer** for AMS Southeastern Sectional Meeting at UTC in Oct. 2022

Journal/Conference Referee

Journal of Statistical Computation and Simulation
 Journal of Biopharmaceutical Statistics
 Journal of Physics A: Mathematical and Theoretical
 Journal of Risk and Financial Management
 Mathematics
 Econometrics
 Stats
 International Journal of Environmental Research and Public Health
 Applied Sciences
 Symmetry
 Entropy
 Perceptual and Motor Skills
 Applied Mathematics and Computation
 Frontiers in Artificial Intelligence
 ACM Mid-Southeast Conference (2021)

Service

Committees

- Community
 - **Secretary** American Statistical Association Eastern Tennessee Chapter (2023 - present)
- University
 - **Committee member** in Course learning evaluations (2024 - 2025)
 - **Member** in Faculty Senate (2023- 2025)
- College
 - **Committee member** in College of Arts & Sciences Curriculum Committee (2022 - 2024)
- Department
 - **Committee Chair**
Academic Awards Committee (2024 - present)

– **Committee member**

Undergraduate Committee (2024 - present) Advisory Committee (2023- 2024)

Graduate Committee (2023- present)

Budget Committee (2022 - 2024, 2025 - 2026)

Academic Awards Committee (2021 - 2024)

Math Competitions Committee (2021 - 2022)

Volunteers

- **Volunteer monitor** for 2025 JSM Continuing Education course
- **Volunteer proctor** for 2024 Math Kangaroo
- **Volunteer grader** for 2024 TMTA Math Competition
- **Volunteer** for 2024 TVR Middle School Drone Competition Championship
- **Volunteer Judge for MAA Student Poster Session - Applied Mathematics Session**
during Joint Mathematics Meetings 2021